

Diagnostic Evaluation Report

1. Framework Information

- **Framework:** X-RAG Diagnostic Framework (v1.0)
- **Trace ID:** `e374c848-4308-4c51-8636-d743fc62c034`
- **Timestamp:** 2026-09-05T14:40:50.793983Z
- **Status:** COMPLETED

2. Executive Summary

- **Question:** What was decided in [2023] 8 S.C.R. 152?
- **Answer:** I do not have enough information to answer this.
- **Health Score:** 0.65
- **Primary Issue:** UNKNOWN
- **Claims Supported:** 1 / 1
- **Summary:** Analysis completed.

3. Pipeline Overview

Stage	Status	Confidence	Observation
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CORPUS	PASS	0.9	Corpus contains sufficient relevant content, evidenced by at least one supported claim.
RETRIEVER	PASS	0.95	At least one retrieved chunk provides supporting evidence for a verified claim.
GENERATOR	PASS	0.95	All generated claims are fully supported by evidence.
GROUNDING	PASS	0.9	No claims are contradicted by retrieved evidence.

Efficiency: 1/6 chunks used (17%)

4. Evaluation Metrics

- **Retrieved Chunks:** 6
- **Verified Claims:** 1 (Supported: 1, Partial: 0, Unsupported: 0, Contradicted: 0)
- **Grounding Score:** 1.00
- **Evidence Coverage:** 0.17
- **Average Entailment:** 0.98
- **Latencies:** Retrieval (2431.5ms) | Generation (13659.5ms) | Verification (4806.8ms)

5. RAGAS Metrics

- **Faithfulness:** 1.0000
- **Answer Relevancy:** 0.4223
- **Context Precision:** 0.0000
- **Context Relevancy:** 0.0000
- **Context Recall:** N/A
- **Answer Similarity:** N/A
- **Answer Correctness:** N/A

Answer Correctness (Claim Recall)

Not computed for this trace.

6. Evidence Analysis

- **[b2fcc9d-0e00-4de3-869d-88b8815d9a41]** (SUPPORTED): I do not have enough information to answer this.
- *Evidence (Chunk f2564119-6af6-4a1b-8f21-a5077975e571):* (d) A witness, of whom nothing whatever is known, being questioned of life and means of living, gives unsatisfactory answers.

7. Root Cause Analysis

- **Primary Cause:** UNKNOWN
- **Confidence:** 0.93
- **Reasoning:**
 - Starting root cause analysis for trace e374c848-4308-4c51-8636-d743fc62c034.
 - CORPUS passed (confidence=0.90).
 - RETRIEVER passed (confidence=0.95).
 - Skipping CHUNKING: no observable state in matrix.
 - GENERATOR passed (confidence=0.95).
 - GROUNDING passed (confidence=0.90).
 - No failures detected. Pipeline is healthy or all evidence is UNKNOWN.

8. Corrective Action Plan

- **[INFORMATIONAL] Review Retrieval Top-K / Reranker Cutoff**
- Description: Retrieval efficiency is low (17%); consider reducing top-k or tightening the reranker score cutoff.
- Expected Improvement: Reduced prompt size and lower latency/cost with no loss of verified support. (Metric: Chunk utilization)

9. Evidence Traceability

- **Primary Cause:** UNKNOWN
- **Derived From:** RootCauseAnalysis
- **Pipeline Stage:** UNKNOWN
- **Supporting Claims:** b2fcce9d-0e00-4de3-869d-88b8815d9a41
- **Supporting Chunks:** f2564119-6af6-4a1b-8f21-a5077975e571

10. Overall Assessment

- **Major Strength:** RETRIEVER is healthy (confidence=95%): At least one retrieved chunk provides supporting evidence for
- **Major Weakness:** UNKNOWN
- **Next Priority:** Review Retrieval Top-K / Reranker Cutoff
- **Recommendation:** Pipeline requires targeted improvements to the primary failure stage.

11. Framework Footer

- **Generated By:** X-RAG Diagnostic Framework
- **Artifact Version:** 1.0

APPENDIX

Artifacts Generated

✓ RAGTrace

artifacts/rag_traces/e374c848-4308-4c51-8636-d743fc62c034.json

✓ ClaimSet

artifacts/claim_sets/e374c848-4308-4c51-8636-d743fc62c034.json

✓ VerificationResults

artifacts/verification/e374c848-4308-4c51-8636-d743fc62c034.json

✓ PipelineStateMatrix

artifacts/pipeline_state_matrix/e374c848-4308-4c51-8636-d743fc62c034.json

✓ RootCauseAnalysis

artifacts/root_cause_analysis/e374c848-4308-4c51-8636-d743fc62c034.json

✓ CorrectiveActionPlan

artifacts/corrective_action_plan/e374c848-4308-4c51-8636-d743fc62c034.json

✓ DiagnosticEvaluationReport

artifacts/reports/e374c848-4308-4c51-8636-d743fc62c034.json